CCNA

VLAN and INTER-VLAN Routing TASK – 4

1. **Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.**

Ans :

```
PC-A ─ Switch ─ R1 ─────── WAN ─────── R2 ─ Switch ─ PC-
B

          \              /

           \--- GRE VPN ---/

                Tunnel 0


        Internet / External Server
```

R1 Commands

- enable
- configure terminal
- interface gigabitEthernet 0/0
- ip address 192.168.1.1 255.255.255.0
- ip nat inside
- no shutdown
- exit
- interface gigabitEthernet 0/1
- ip address 10.0.0.1 255.255.255.252
- ip nat outside
- no shutdown
- exit

- interface tunnel 0
- ip address 172.16.0.1 255.255.255.252
- tunnel source gigabitEthernet 0/1
- tunnel destination 10.0.0.2
- no shutdown
- exit
- access-list 1 permit 192.168.1.0 0.0.0.255
- ip nat inside source list 1 interface gigabitEthernet 0/1 overload
- ip route 192.168.2.0 255.255.255.0 172.16.0.2
- ip route 0.0.0.0 0.0.0.0 10.0.0.2
- end

R2 Commands

- enable
- configure terminal
- interface gigabitEthernet 0/0
- ip address 192.168.2.1 255.255.255.0
- ip nat inside
- no shutdown
- exit
- interface gigabitEthernet 0/1
- ip address 10.0.0.2 255.255.255.252
- ip nat outside
- no shutdown
- exit
- interface tunnel 0
- ip address 172.16.0.2 255.255.255.252
- tunnel source gigabitEthernet 0/1
- tunnel destination 10.0.0.1

- no shutdown
- exit
- access-list 1 permit 192.168.2.0 0.0.0.255
- ip nat inside source list 1 interface gigabitEthernet 0/1 overload
- ip route 192.168.1.0 255.255.255.0 172.16.0.1
- ip route 0.0.0.0 0.0.0.0 10.0.0.1
- end

2. **Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address.

Hint: Use the 'overload' keyword in your NAT configuration.**

   Ans :

   - enable
   - configure terminal
   - interface gigabitEthernet 0/0
   - ip address 192.168.1.1 255.255.255.0
   - ip nat inside
   - no shutdown
   - exit
   - interface gigabitEthernet 0/1
   - ip address 10.0.0.1 255.255.255.252
   - ip nat outside
   - no shutdown
   - exit
   - access-list 1 permit 192.168.1.0 0.0.0.255
   - ip nat inside source list 1 interface gigabitEthernet 0/1 overload
   - ip route 0.0.0.0 0.0.0.0 10.0.0.2
   - end

- show ip nat translations
- show ip nat statistics

3. **Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.**

   Ans :

   - R1:
   - enable
   - configure terminal
   - crypto isakmp key wrongkey address 10.0.0.2
   - end
   - show crypto isakmp sa
   - show running-config
   - configure terminal
   - no crypto isakmp key wrongkey address 10.0.0.2
   - crypto isakmp key cisco123 address 10.0.0.2
   - end
   - show crypto isakmp sa
   - show running-config
   - ping 172.16.0.2

   - R2:
   - enable
   - configure terminal
   - crypto isakmp key cisco123 address 10.0.0.1
   - end
   - show crypto isakmp sa
   - show running-config
   - ping 172.16.0.1

4. **Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.**

   Ans :

   - R1:
   - enable
   - show interfaces tunnel 0
   - ping 172.16.0.2
   - show ip route
   - configure terminal
   - ip route 192.168.2.0 255.255.255.0 172.16.0.2
   - end
   - show ip route
   - ping 192.168.2.10

   - R2:
   - enable
   - show interfaces tunnel 0
   - ping 172.16.0.1
   - show ip route
   - configure terminal
   - ip route 192.168.1.0 255.255.255.0 172.16.0.1
   - end
   - show ip route
   - ping 192.168.1.10

5. **Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it.

Constraint: Focus on using**

'show' and 'debug' commands for troubleshooting rather than guessing the issue.

Ans : After receiving my classmate's WAN, VPN, and NAT Packet Tracer file, I first checked the router interfaces using show ip interface brief and verified the routing table with show ip route. I then checked the VPN status using show interfaces tunnel 0 and show crypto isakmp sa. The VPN tunnel was not establishing correctly. To investigate further, I used debug crypto isakmp, which showed that the pre-shared keys between the two routers did not match. I used show running-config to confirm the incorrect key in the configuration. I corrected the key so that both routers used the same pre-shared key, then disabled debugging with undebug all. Finally, I tested the connection using ping and confirmed that the VPN tunnel and branch-to-branch connectivity were working.